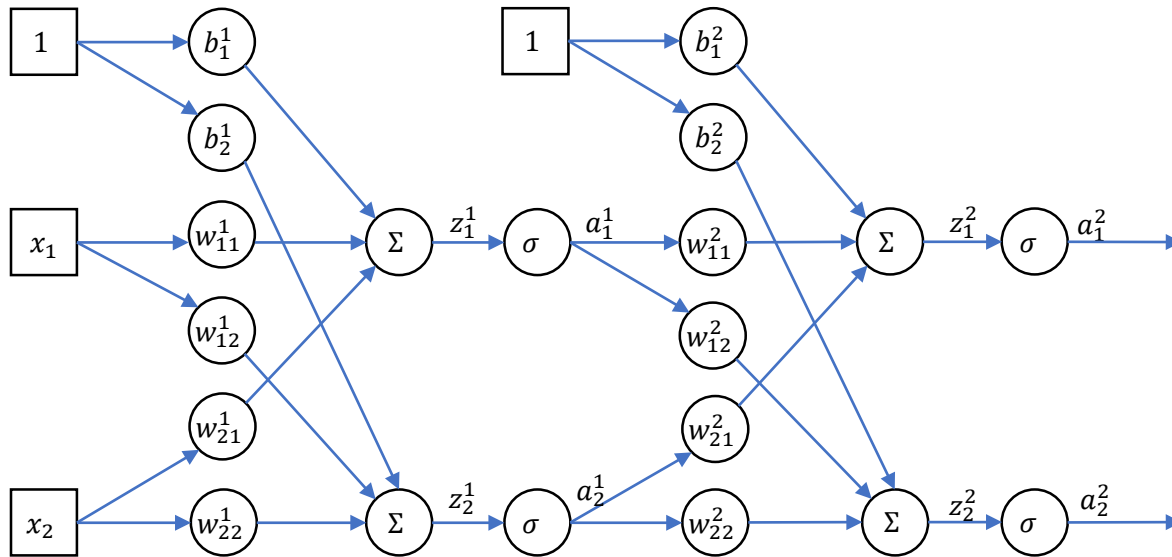


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Forward pass example



Forward pass



$$\mathbf{z}^1 = (W^1)^T \cdot \mathbf{x} + \mathbf{b}^1$$

$$\mathbf{a}^1 = \sigma(\mathbf{z}^1) = \begin{bmatrix} \sigma(z_1^1) \\ \sigma(z_2^1) \end{bmatrix}$$

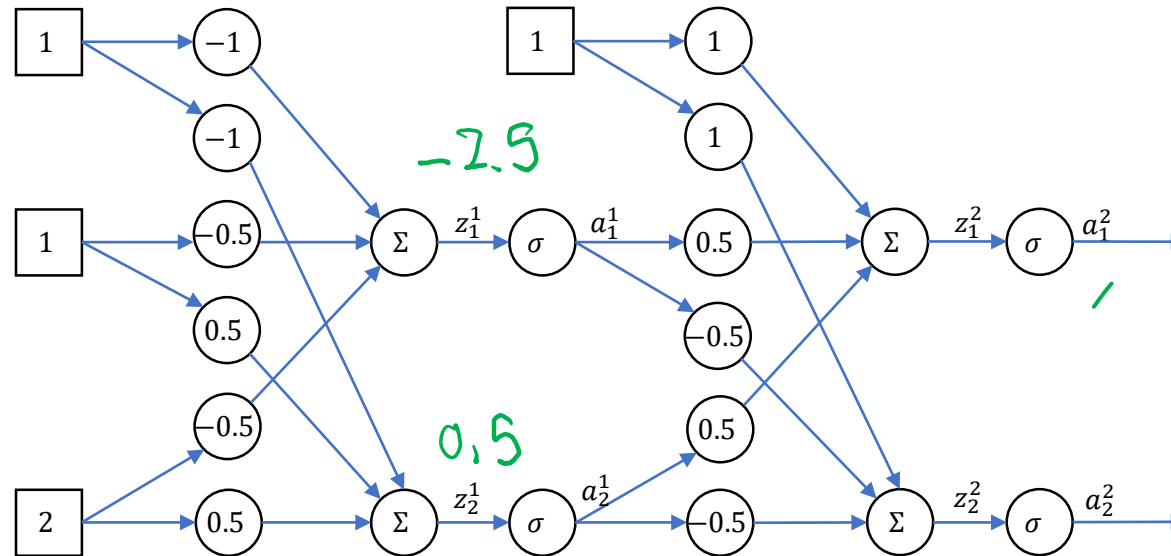
$$\mathbf{z}^2 = (W^2)^T \cdot \mathbf{a}^1 + \mathbf{b}^2$$

$$\mathbf{a}^2 = \sigma(\mathbf{z}^2) = \begin{bmatrix} \sigma(z_1^2) \\ \sigma(z_2^2) \end{bmatrix}$$



Numerical Example - ReLU

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{b}^1 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \mathbf{b}^2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, W^1 = \begin{bmatrix} -0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix}, W^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0.5 & -0.5 \end{bmatrix}$$



$$\sigma(z) = \text{ReLU}(z) = \max(0, z)$$

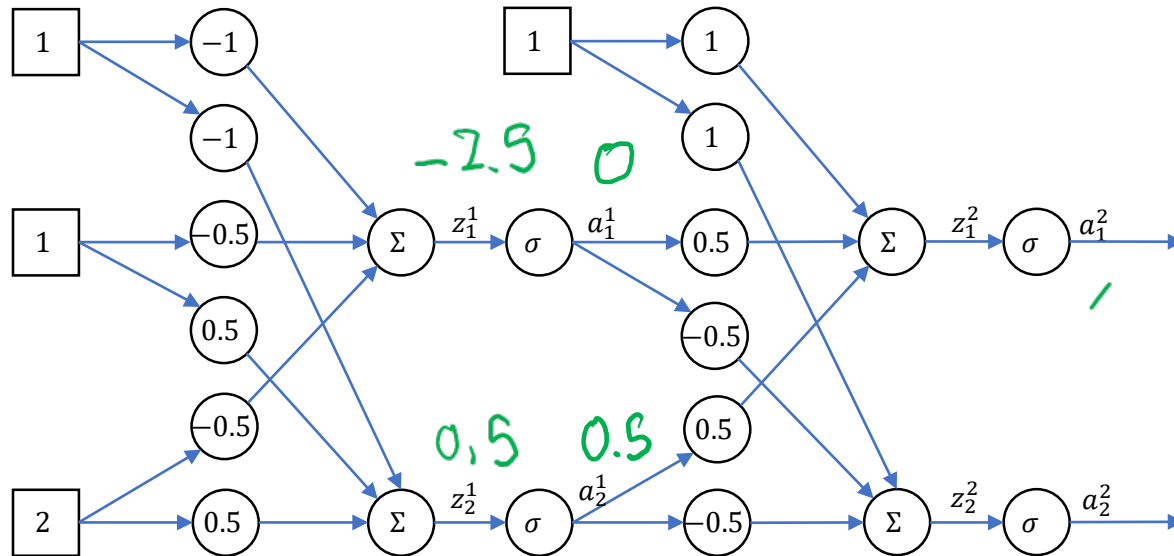


$$\begin{aligned} \mathbf{z}^1 &= (W^1)^T \cdot \mathbf{x} + \mathbf{b}^1 \\ &= \begin{bmatrix} -0.5 & -0.5 \\ 0.5 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} -2.5 \\ 0.5 \end{bmatrix} \end{aligned}$$



Numerical Example - ReLU

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{b}^1 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \mathbf{b}^2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, W^1 = \begin{bmatrix} -0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix}, W^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0.5 & -0.5 \end{bmatrix}$$



$$\sigma(z) = \text{ReLU}(z) = \max(0, z)$$

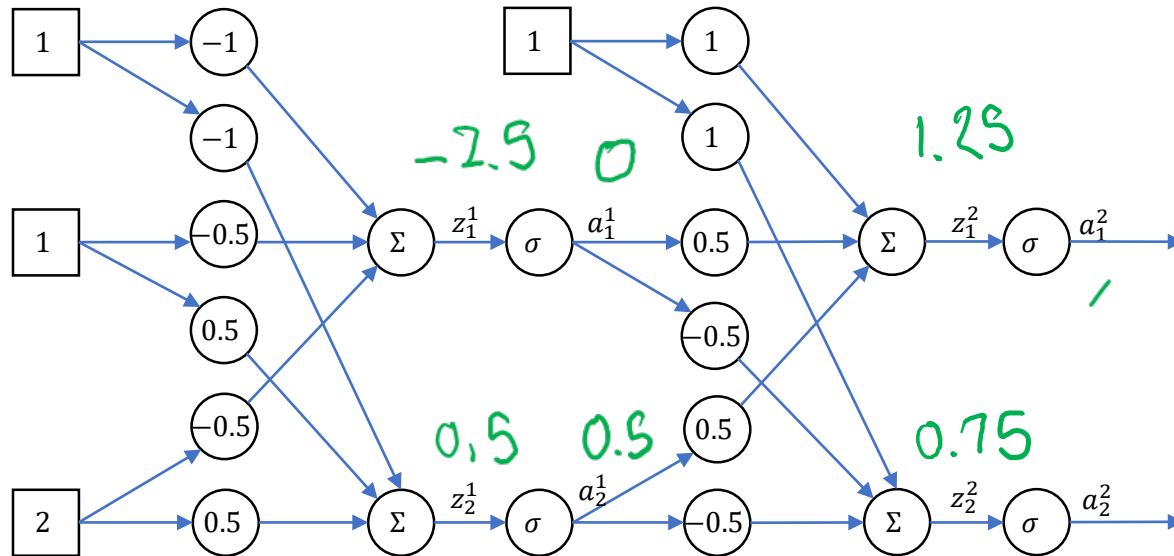


$$\begin{aligned} \mathbf{z}^1 &= (W^1)^T \cdot \mathbf{x} + \mathbf{b}^1 \\ &= \begin{bmatrix} -0.5 & -0.5 \\ 0.5 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} -2.5 \\ 0.5 \end{bmatrix} \\ \mathbf{a}^1 &= \sigma(\mathbf{z}^1) \\ &= \begin{bmatrix} \sigma(z_1^1) \\ \sigma(z_2^1) \end{bmatrix} = \begin{bmatrix} \max(0, z_1^1) \\ \max(0, z_2^1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} \end{aligned}$$



Numerical Example - ReLU

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{b}^1 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \mathbf{b}^2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, W^1 = \begin{bmatrix} -0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix}, W^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0.5 & -0.5 \end{bmatrix}$$



$$\sigma(z) = \text{ReLU}(z) = \max(0, z)$$

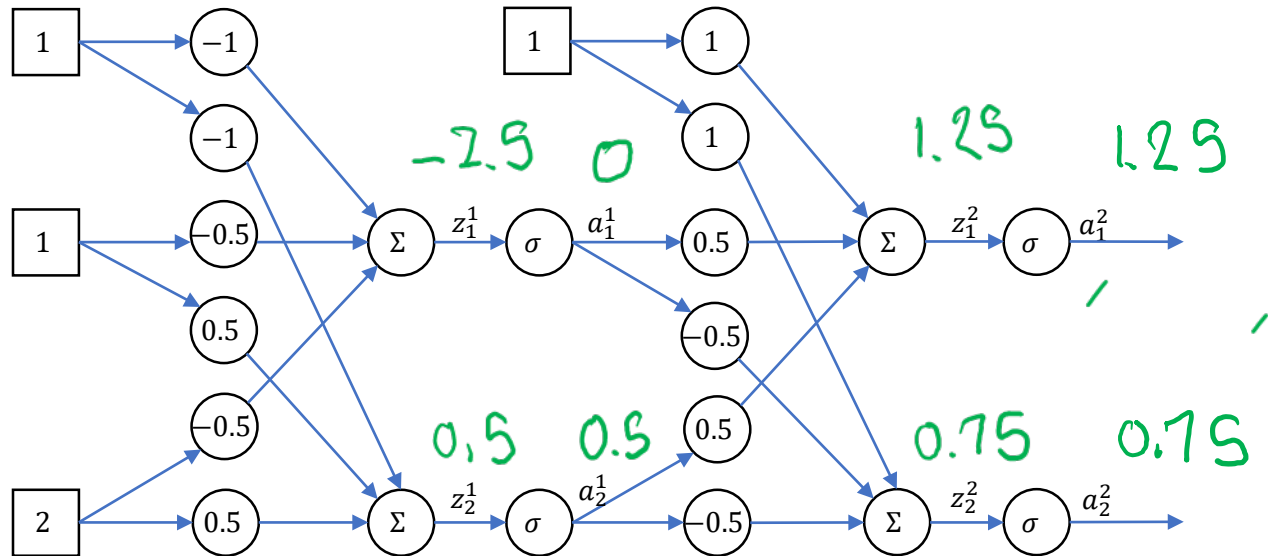


$$\begin{aligned} \mathbf{z}^1 &= (W^1)^T \cdot \mathbf{x} + \mathbf{b}^1 \\ &= \begin{bmatrix} -0.5 & -0.5 \\ 0.5 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} -2.5 \\ 0.5 \end{bmatrix} \\ \mathbf{a}^1 &= \sigma(\mathbf{z}^1) \\ &= \begin{bmatrix} \sigma(z_1^1) \\ \sigma(z_2^1) \end{bmatrix} = \begin{bmatrix} \max(0, z_1^1) \\ \max(0, z_2^1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} \\ \mathbf{z}^2 &= (W^2)^T \cdot \mathbf{a}^1 + \mathbf{b}^2 \\ &= \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & -0.5 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1.25 \\ 0.75 \end{bmatrix} \end{aligned}$$



Numerical Example - ReLU

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{b}^1 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \mathbf{b}^2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, W^1 = \begin{bmatrix} -0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix}, W^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0.5 & -0.5 \end{bmatrix}$$



$$\sigma(z) = \text{ReLU}(z) = \max(0, z)$$



$$\begin{aligned} \mathbf{z}^1 &= (W^1)^T \cdot \mathbf{x} + \mathbf{b}^1 \\ &= \begin{bmatrix} -0.5 & -0.5 \\ 0.5 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} -2.5 \\ 0.5 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \mathbf{a}^1 &= \sigma(\mathbf{z}^1) \\ &= \begin{bmatrix} \sigma(z_1^1) \\ \sigma(z_2^1) \end{bmatrix} = \begin{bmatrix} \max(0, z_1^1) \\ \max(0, z_2^1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \mathbf{z}^2 &= (W^2)^T \cdot \mathbf{a}^1 + \mathbf{b}^2 \\ &= \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & -0.5 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1.25 \\ 0.75 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \mathbf{a}^2 &= \sigma(\mathbf{z}^2) \\ &= \begin{bmatrix} \sigma(z_1^2) \\ \sigma(z_2^2) \end{bmatrix} = \begin{bmatrix} \max(0, z_1^2) \\ \max(0, z_2^2) \end{bmatrix} = \begin{bmatrix} 1.25 \\ 0.75 \end{bmatrix} \end{aligned}$$



Next up:

Backpropagation

